ASSIGNMENT NO .2

Problem 1: Grade Evaluation System

Problem Statement:
Write a Java program that calculates the average marks of a student and determines the grade based on the following criteria:

Grade A: Average marks ≥ 90
Grade B: Average marks between 70 and 89
Grade C: Average marks between 50 and 69
Grade D: Average marks between 30 and 49
Fail: Average marks < 30

Code:

```
public class GradeEvaluation  {
   public static void main(String[] args) {
      int maths = 80;
      int science = 85;
      int history = 90;
      int average = (maths + science + history) / 3;
      String grade;
      if (average >= 90) {
         grade = "A";
      } else if (average >= 70) {
         grade = "B";
      } else if (average >= 50) {
         grade = "C";
      } else if (average >= 30) {
         grade = "D";
      } else {
         grade = "Fail";
      }
      System.out.println("Average Marks: " + (int) average);
      System.out.println("Grade: " + grade);
   }
}
```

Expected Output:

Average Marks: 85
Grade: B

Problem 2: Leap Year

Problem Statement:

Write a Java program that checks whether the year is a leap year or not. A year is a leap year if:
It is divisible by 4, but not divisible by 100, or
It is divisible by 400.

Code:

```
import java.util.Scanner;
public class LeapYear {
    public static void main(String[] args) {
        int year =  2024;
        boolean isLeap = (year % 4 == 0 && year % 100 != 0) || (year % 400 == 0);
        if (isLeap) {
            System.out.println(year + " is a leap year.");
        } else {
            System.out.println(year + " is not a leap year.");
        }
    }
}
```

Expected Output:

2024 is a leap year.

Problem 3: Days of the Week

Problem Statement:

Write a Java program that takes an integer between 1 and 7 and prints the corresponding day of the week using a switch-case statement. If the input is outside the range of 1 to 7, the program should display "Invalid day number".

Code:

```
import java.util.Scanner;
public class DayNumber {
    public static void main(String[] args) {
        int dayNumber = 3;

        switch (dayNumber) {
            case 1:
                System.out.println("The day is Monday");
                break;
            case 2:
                System.out.println("The day is Tuesday");
                break;
```

```
        case 3:
            System.out.println("The day is Wednesday");
            break;
        case 4:
            System.out.println("The day is Thursday");
            break;
        case 5:
            System.out.println("The day is Friday");
            break;
        case 6:
            System.out.println("The day is Saturday");
            break;
        case 7:
            System.out.println("The day is Sunday");
            break;
        default:
            System.out.println("Invalid day number");
        }
    }
}
```

Expected Output:

The day is Wednesday

Problem 4: Identify the Values of Uninitialized Variables

Scenario:

You are working on a program that handles different data types. Your manager has asked you to quickly check the values of various variables, but you're in a rush and forget to initialize them. As you go through the code, you expect some values to show up, but Java has something else in mind. Your task is to fix the issue and ensure the variables hold proper values.

Instructions:

1.Declare the following variables:

byte a; short b; int c; long d; float e; double f; char g; boolean h;

2.Print out their values.

Code:

```
public class UninitializedVariables {
    public static void main(String[] args) {
        byte a = 10;
        short b = 200;
        int c = 100000;
        long d = 10000000000L;
        float e = 3.14f;
        double f = 3.14159;
        char g = 'A';
        boolean h = true;

        System.out.println("a = " + a);
        System.out.println("b = " + b);
        System.out.println("c = " + c);
        System.out.println("d = " + d);
        System.out.println("e = " + e);
        System.out.println("f = " + f);
        System.out.println("g = " + g);
        System.out.println("h = " + h);
    }
}
```

output:

```
a = 10
b = 200
c = 100000
d = 10000000000
e = 3.14
f = 3.14159
g = A
h = true
```